

Climate Science, UNFCCC & Paris Agreement

GHG physics, UNFCCC - Kyoto - Paris arc, NDCs/COP and India pledges - rising #1 environment portion.

Portion focus ~88% · Topic: env-climate

PYQ themes: Paris Agreement · NDCs · COP · GHGs · Mitigation/adaptation

Why this portion (PYQ lens)

Ranked among the highest next-exam-chance portions (~88%, rising). Appeared across 2016 - 2025; late-window COP/NDC weight exceeds early years. Parent topic env-climate ~91%.

Years appeared (synthesis): 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025

Why UPSC asks this

Blends durable science with living treaty politics and India's diplomacy - ideal for mixed static - CA papers.

Syllabus framing

Ranked among the highest next-exam-chance portions (~88%, rising). Appeared across 2016 - 2025; late-window COP/NDC weight exceeds early years. Parent topic env-climate ~91%. This portion sits inside env-climate and centres on climate science and the UNFCCC/Paris regime. Blends durable science with living treaty politics and India's diplomacy - ideal for mixed static - CA papers.

Recurring PYQ angles: Paris Agreement, NDCs, COP, GHGs, Mitigation/adaptation. Approximate appearance years in the 2016-2025 educational synthesis: 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025. Use the chance percentage only to sequence study hours - it is not an official prediction.

Core map of climate science and the UNFCCC/Paris regime

Enhanced greenhouse effect: Anthropogenic GHGs increase infrared absorption in the troposphere, raising global mean temperatures beyond the natural greenhouse baseline. Major GHGs: CO₂ (long-lived), CH₄ (higher GWP, shorter life), N₂O, and F-gases.

GWP: Compares warming of a gas to CO₂ over a chosen time horizon. Carbon sinks: Oceans, forests and soils remove CO₂; not infinite offsets.

Hold this core map as your elimination spine before memorising peripheral CA overlays.

Core concepts and definitions

UNFCCC 1992: Framework convention creating COP process and CBDR-RC. Kyoto Protocol: Historic Annex-I style quantified targets with flexibility mechanisms.

Paris Agreement 2015: Universal NDCs; temperature goals well below 2C / pursue 1.5C. Global Stocktake: Collective progress check to inform stronger NDCs.

Institutions, processes and mechanisms

Transparency framework: Reporting/review rules building trust. Mitigation: Actions that reduce emissions or enhance sinks.

Adaptation: Adjustments to limit harm from climate impacts. Loss and damage: Agenda on residual irreversible harms.

India angle and applied examples

CBDR-RC: Common but differentiated responsibilities and respective capabilities. India narrative: Equity, intensity and non-fossil pledges, net-zero framing - verify latest NDC numbers.

Montreal contrast: Ozone treaty - do not merge into UNFCCC. Co-benefits: Air quality and energy security gains from climate action.

Comparisons, linkages and discriminators

Extremes literacy: Warming shifts probabilities of heat extremes. Ocean acidification: CO₂ dissolution alters carbonate chemistry.

COP theatre: Revise last 2 - 3 COP headlines as CA overlay. Carbon budget idea: Finite CO₂ consistent with a temperature target.

Science spine for Prelims

The enhanced greenhouse effect describes how anthropogenic greenhouse gases increase infrared absorption in the troposphere, raising global mean temperatures above the natural greenhouse baseline. CO₂ is long-lived and central to fossil and land-use stories; CH₄ has higher global warming potential but a shorter atmospheric lifetime; N₂O and industrial F-gases complete the usual set. Global Warming Potential compares gases to CO₂ over a chosen horizon (often 100 years in textbooks).

Carbon sinks in oceans, forests and soils remove CO₂ on different timescales - they are not infinite offsets for endless emissions. Ocean acidification from CO₂ dissolution alters carbonate chemistry and is linked to, but distinct from, warming alone. Extremes literacy matters: warming shifts probabilities of heat extremes and influences heavy rainfall statistics without making every single storm a simple fingerprint slogan.

Treaty arc: UNFCCC to Paris

UNFCCC (1992) created the COP process and the equity principle of common but differentiated responsibilities and respective capabilities (CBDR-RC). Kyoto is remembered for Annex-I style quantified targets and flexibility mechanisms in the classic narrative. Paris (2015) shifted to near-universal Nationally Determined Contributions with temperature goals of well below 2 deg C and pursuit of 1.5 deg C, plus a Global Stocktake and transparency framework meant to ratchet ambition.

Do not treat Kyoto and Paris as legally identical designs. Do not merge the Montreal Protocol (ozone-depleting substances) into UNFCCC casually. Mitigation cuts emissions or enhances sinks; adaptation limits harm from impacts already in the pipeline; loss and damage politics address residual irreversible harms for vulnerable states.

India angle

India emphasises equity and development space while expanding renewables and long-term net-zero narratives. Keep three pledge anchors updated from official NDC documents.

Question patterns and exam spine

Treaty design statements; GHG/sinks science; COP finance language; India pledges.

Revision spine: GHG science -> UNFCCC/Kyoto/Paris -> NDC/stocktake -> India anchors -> Montreal trap.

Eliminate anachronisms and category errors first, then test remaining options against table rows and must-remember triggers.

Trap watch-list (specific)

Avoid: Treating Kyoto and Paris as identical. Avoid: Confusing Montreal with UNFCCC. Avoid: NDCs as mere press releases. Avoid: Net zero as overnight zero emissions. Avoid: Collapsing mitigation/adaptation/L&D. Avoid: Infinite forest offsets.

Treaty design contrast

Feature | Kyoto (classic) | Paris

Participation -- Annex-I focus historically -- Near-universal NDCs

Target style -- Top-down quantified -- Bottom-up NDCs

Temperature goal -- Not Paris-style 1.5/2 framing -- Well below 2C; pursue 1.5C

Ambition cycle -- Commitment periods -- NDC cycles + Stocktake

Vocabulary triad

Term | Focus | Prelims cue

Mitigation -- Cut/sequester GHGs -- Renewables, sinks

Adaptation -- Live with impacts -- EWS, resilient infra

Loss & damage -- Residual harm finance/politics -- Vulnerable nations

Map / list facts

- Himalayan cryosphere
- Coastal vulnerability
- SIDS L&D politics
- India renewables geography

Common traps

- x Treating Kyoto and Paris as identical
- x Confusing Montreal with UNFCCC
- x NDCs as mere press releases
- x Net zero as overnight zero emissions
- x Collapsing mitigation/adaptation/L&D
- x Infinite forest offsets

Must remember

- UNFCCC -> Kyoto -> Paris differences
- GHGs + GWP idea
- Paris NDCs/stocktake
- CBDR-RC
- Mitigation vs adaptation vs L&D
- Refresh India NDC numbers
- Montreal != UNFCCC

Quick revision checklist

- [] Redraw Kyoto vs Paris table
- [] List GHG + sink
- [] Define mitigation/adaptation/L&D
- [] Update India pledges

[] Last 2 COP bullets

Educational prioritisation heuristic - not an official UPSC prediction. (c) UPSC IAS Study Hub original notes.